THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 1: **Exploring the Investigative World of Science** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A pendulum completes 30 swings in 48 s. The time period is: 1

(a) 1.6 s **(b)** 0.625 s **(c)** 16 s **(d)** 78 s

2. Which is the best question to start a study of how the volume of water affects the time to boil? 1

(a) Who boils water?
(b) Does 200 mL of water boil faster than 400 mL on the same burner?
(c) Where do we boil water?
(d) Is boiled water good?

3. Assertion (A): The average of several readings is more reliable than a single reading. Reason (R): Random errors tend to cancel out when readings are averaged. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. Mean of 2.0, 2.2, 2.1, 5.0 and 2.1 is 2.68. Excluding the outlier 5.0 gives mean: 1

(a) 2.0 **(b)** 3.0 **(c)** 2.68 **(d)** 2.1

5. Different groups plot their data on a graph. Points of one group lie far from the others. First step is to: 1

(a) delete the group **(b)** average all
(c) ignore the graph **(d)** check the procedure of that group

6. Assertion (A): A thermometer reading is a quantitative observation. Reason (R): It is expressed as a number with a unit. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. A claim: 'plants grow best near the window because of light'. A weakness of the claim is: 1

(a) windows are glass **(b)** light is not needed
(c) plants cannot grow **(d)** temperature and water were not compared

8. If a measurement is written as 2.5 without a unit, the problem is: 1

(a) it is an inference **(b)** it is accurate
(c) it is not meaningful **(d)** it is a conclusion

9. **Assertion (A):** A single measurement is enough to be sure of a result. 1
Reason (R): Repeating a measurement and taking the average reduces the effect of random errors.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Reading a scale with the eye placed to the side of the pointer gives an incorrect value. 1
Reason (R): This error can be removed by using a thicker scale.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Identify the flaw: 'We tested the heat conduction of rods using different amounts of wax on each.' 2

12. Why do scientists give the method in detail in a report? 2

13. A plant grew 3 cm in dark and 8 cm in light in a week. State the hypothesis supported and one limitation. 2

14. Why is a 'dummy' set without the treatment kept in medicine trials? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how the principle of fair testing is applied while comparing two brands of detergent for removing stains. 3

16. A friend says 'My experiment failed because it did not give the expected result.' Reply scientifically. 3

17. Explain why one can be misled by a graph with a truncated (cut) axis. 3

OR

List the variables in the investigation 'Does the size of a candle flame affect how fast water heats up?' and state a suitable hypothesis. 3

18. Explain the importance of peer review (sharing findings with other scientists). 3

OR

Why do scientists use symbols and standard units when reporting results? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A student records the temperature of hot tea every 2 minutes: 0 min - 90 degrees C, 2 min - 78 degrees C, 4 min - 69 degrees C, 6 min - 62 degrees C. 5
(i) How much did the temperature fall in the first 2 minutes?
(ii) How much did it fall between 4 and 6 minutes?
(iii) Which instrument measures the temperature?
(iv) Does the tea cool at the same rate throughout? Justify using the data.

20. Evaluate this investigation: 'To see if music helps plants grow, Ravi played music to a plant in his room and compared it with a plant in his friend's room.' 5

OR

Distinguish between a scientific law, a hypothesis and an opinion, with examples. 5

END OF QUESTION PAPER — Total: 40 Marks